

Crash Hotspot Detection in Dubai using DBSCAN and Severity Weighting

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Abstract—Rapidly growing cities as Dubai, frequently experience high number of incidents related to traffic. To reduce those incidents, transportation authorities must prioritize and classify locations with high number of accidents. This research introduces a method for identifying high-crash areas, employing DBSCAN clustering and severity-weighted hotspot scoring. The methodology utilizes geographic coordinates, road classifications, and community designations to improve the clarity of the results. The analysis showed that certain areas, particularly major roads, had a higher number of accidents. The model is proving to be quite useful, easily understood, and it fits right in with smart mobility applications.

Index Terms—Crash Hotspots, DBSCAN, Smart Mobility, Spatial Analysis, Traffic Safety, Dubai

I. INTRODUCTION

Dubai has experienced rapid growth, where more people moved to live in the city. As a result, the number of people driving cars have increased and road safety became a concern. Finding crash hotspots, or places where there are a lot of traffic accidents, is one of the biggest problems with studying traffic safety.

Standard methods don't always work with traffic data that has spatial dependencies. Accidents happen more often at intersections, on major roads, and in places where many people live close together. So, spatial data mining techniques are more effective for identifying these patterns. DBSCAN is a density-based clustering algorithm capable of identifying arbitrarily shaped clusters while separating noise points [1], [2].

This study provides an approach to identify and rank spots where there can be a high number of crashes. It includes spatial clustering, risk assessment, and analysis of crash patterns over time. The framework is a key advantage for making decisions, as it is able to recognize dynamic hotspots using map-based reverse geocoding.

II. DATASET DESCRIPTION

The analysis is based on a structured traffic incident dataset obtained from the Dubai open-data ecosystem[3]. The dataset's structure mirrors that of the Dubai Police's traffic incident documentation. Each record within the dataset encompasses multiple data fields: an accident identifier, a description of the incident presented in Arabic, the geographical coordinates (longitude and latitude), the precise time of the accident, and the timestamp indicating when the data was loaded. This study uses a dataset with around 5,000 records, all captured between March 14 and March 30, 2026.

Data preprocessing step required removing records with missing or incorrect coordinate data. Moreover, it involved filtering the observations to ensure they are within the geographical boundaries of Dubai. Temporal analysis necessitated converting timestamp fields into a datetime format. When a dedicated severity variable is unavailable, accident category text serves as the basis for determining severity levels through keyword matching. Minor incidents were assigned a weight of 1, moderate incidents a weight of 2, and severe incidents a weight of 3. To ensure comprehensive data inclusion, all other categories were allocated a default weight of 1.

III. METHODOLOGY

The method represent each accident as a point using the latitude and longitude. The DBSCAN clustering algorithm looks at the spatial distribution of incidents by grouping points based on how dense they are, not by how they are shaped.

DBSCAN has two settings: a neighborhood radius ϵ and a minimum number of points, *MinPts*. The radius, measured in meters, is used in the Haversine distance metric, which accounts for the Earth's curvature by converting it to radians. Points that are close together are grouped into clusters, while points that are far apart are considered noise. The algorithm does not require specifying the number of clusters and is suited for spatial hotspot detection [1], [2]. Each cluster gets a severity-weighted score to measure the risk of a hotspot. The risk score for a group c is defined as:

$$Risk_c = \sum_{i \in c} w_i \quad (1)$$

Here, w_i represents the severity weight of incident i . This method ensures that clusters with more severe incidents are ranked higher than those with only frequent, less severe incidents.

In addition, time patterns were also examined to find periods of increased risk. Incident timestamps are grouped into two-hour periods. The most common interval within each cluster is then identified as the time of highest risk. Therefore, this method allows the framework to include both the spatial and temporal aspects of crash hotspots.

To improve understanding, each cluster is given a location that can change, using reverse geocoding. After identifying the center of each cluster, the data is used in a map-based geocoding service. The address components retrieved from this service are then used to create a descriptive label. This label

provides information about nearby roads, neighborhoods, or districts. Therefore, the designation of hotspots is based on real geographic locations. Reverse geocoding use map-based services that return nearest geographic location associated with given coordinates [4].

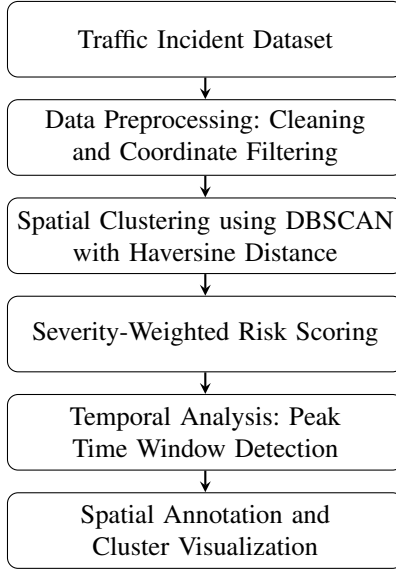


Fig. 1. Proposed framework for crash hotspot detection.

IV. RESULTS

The application of this method produced strong evidence of significant spatial clustering in traffic incidents in Dubai. DBSCAN proved effective, identifying numerous high-density zones that correspond to established crash hotspots, while excluding anomalous data points.

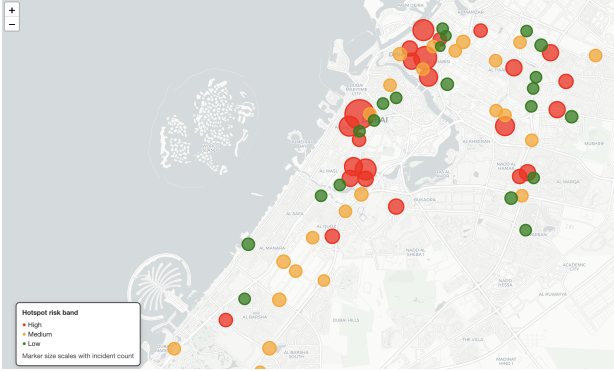


Fig. 2. Spatial distribution of detected crash hotspots.

A scoring system that considers the severity of each incident provides a more informative way to rank hotspots than just counting the number of incidents.

Clusters with fewer but more serious incidents should be prioritized over those with more frequent, less serious events. Analysis of the temporal data shows that each hotspot has a specific period of increased risk. Several of these hotspots

seem to experience a surge in incidents during the evening and late-night hours. This pattern likely stems from heavier traffic and diminished visibility at those times.

Identifying the precise location of each hotspot is essential for understanding them, and reverse geocoding is the method used to accomplish this. The interactive map offers a detailed look at these hotspots. The analysis examines the level of risk, how often incidents occur, their severity, and when they occur.

V. DISCUSSION

The method showed an increase in traffic accidents in specific areas in Dubai. DBSCAN worked accurately in identifying several areas with high accident rates that corresponded with known crash hotspots, while also removing outliers.

The proposed framework delivers a number of advantages. The fact that the number of clusters is not required to be specified is a key benefit. By analyzing the level of severity, it can identify which areas are problematic and require immediate attention. Also, it is able to evaluate the degree of risk. When dynamic naming and analysis are added over time, the results become more practical.

However, there are specific constraints associated with this approach. It is challenging to define road segments within the current clusters, due to non-adequate information on the full road network. Moreover, it is more difficult to determine the precise location of an event when using external providers for reverse geocoding[4]. An incident's perceived seriousness may also be influenced by the data's format.

To reduce the number of incidents potentially, researchers could include more detailed data about road systems, patterns within traffic, weather changes, and other relevant factors. Smart city systems would benefit from a framework that can manage real-time data streams.

VI. CONCLUSION

The objective of the study was to identify areas that have high accidents risk, using DBSCAN clustering, severity-weighted risk scoring, and spatiotemporal analysis. The model works on the identification and prioritization of locations with higher possibilities of accidents. The results support data-driven decision-making for improving road safety and optimizing resource allocation. Including other factors as weather conditions and traffic patterns could improve the model performance.

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